
“Automation applied to an efficient operation will magnify the efficiency.”
— Bill Gates, Business @ the Speed of Thought (1999)

COMPFEST 18
Data Science Academy

Identity-Document OCR for Reliable Structured Identity-Data Extraction



Team Ketolak Pacil



Orie Abyan Maulana



Aditya Akbar



Khansa Firyal Adiila

Identity-Document OCR for Efficient Extraction with Controlled Review

The goal is not to replace human judgment, but to reduce repetitive identity-data extraction while keeping uncertain outputs auditable and reviwable.

1 BUSINESS PROBLEM

Manual identitiy-data extraction is repetitive, time-consuming, and difficult to scale. Heterogenous image quality and document layouts make automation non-trivial.

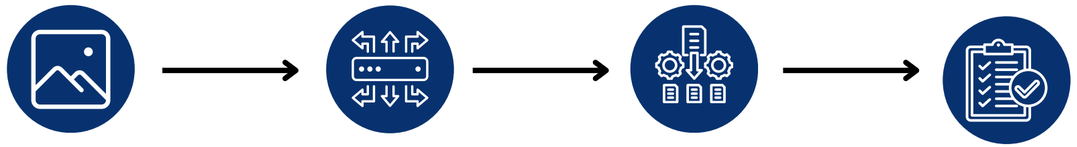
Comparable identity-registration OCR case:
12 → 4 min, 15% → 3% error, 3 x capacity.

Externeal benchmark - Wiranto and Rohim (2026), not our measured impact.

2 WHY AUTOMATION IS HARD

- **Image quality varies** → blur, glare, low light.
- **Documents structure varies** → layouts, families, field conventions.
- **Evidence can be uncertain** → Unknown routes and weak OCR text.

3 OUR TECHNICAL RESPONSE



Adaptive preprocessing, route-specific logic, and field traceability turn noise images into structured fields.

4 RESULT → DEPLOYMENT IMPLICATION

66.77% Macro Field EM

37.66% Full Record EM

81.28% Normalized Similarity

Bottleneck: Name extraction on UNKNOWN_ROUTE fallback cases.

Deployment implication: Automate supported extraction; escalate uncertain cases.

Data Understanding Shapes the Solution Design

The dataset is heterogeneous in image quality, document structure, and repeated captures. Each finding leads to a specific system decision

632 labeled images | **100** test images | **3** target fields: Name, DOB, Address | **152** label-derived groups | **112** Address-present rows, 17.7% of labeled set

1 Image Quality

EVIDENCE



Clean | Blur | Glare | Low Light

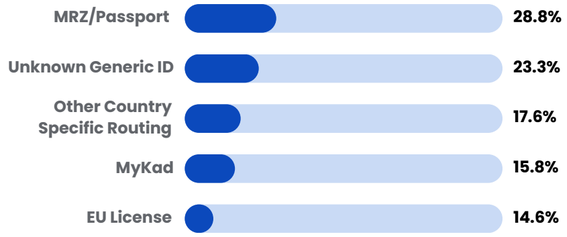
INSIGHT

Image quality varies enough that one preprocessing strategy cannot fit every capture.

2 Document Structure

EVIDENCE

Route distribution (% of labeled images)



INSIGHT

Different layouts and field conventions require different extraction logic.

3 Repeated Captures

EVIDENCE



Capture A — Name: SPECIMEN, DOB: 22.06.1969, Address: -

Capture B — Name: SPECIMEN, DOB: -, Address: -

Capture C — Name: -, DOB: -, Address: -

506/632 labeled images belong to repeated groups | **80.1%**

INSIGHT

Different captures from the same label-derived group may preserve different fields.

EDA FINDING -> SOLUTION DESIGN

1 **Image Quality** → Adaptive preprocessing

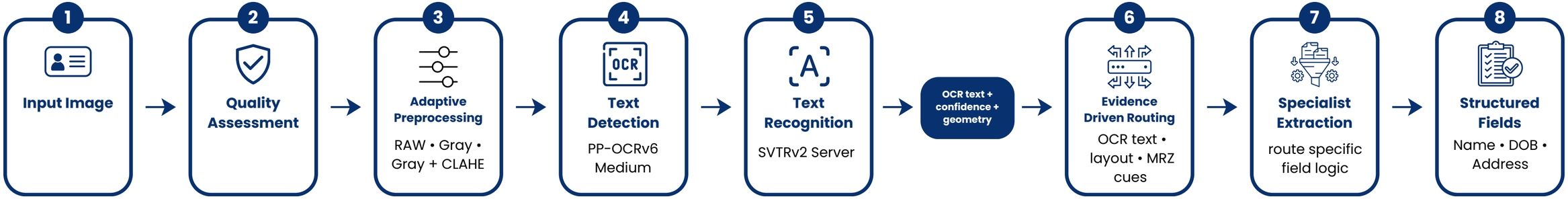
2 **Document Structure** → Evidence Driven Routing + Specialist Extraction

3 **Repeated Captures** → Group Aware Validation + Separate Analysis

Final Architecture Separates Core Extraction From Evaluation Logic

The system converts heterogeneous identity images into structured identity fields, while repeated-capture development and evaluation analyses remain separate from core inference.

CORE OCR PIPELINE



SEPARATE DEVELOPMENT AND EVALUATION LAYER

Group Aware Validation
for supervised components where applicable

Identity Linking Development Experiment
V3 Siamese linking exploration

Downstream Labeled Benchmark Analysis
LOO peer analysis using known label-derived groups

WHY THESE METRICS?

Macro Field EM
averages field-level exact match so Name, DOB, and Address contribute equally

Full Record EM
checks whether all required fields are correct together

Normalized Similarity
captures near-correct strings

Record CER
measures character-level error severity

FROM IMAGE to OCR Evidence

Before any field extraction, the document is converted into text, confidence, and geometry.

ACTUAL EXAMPLE TRACE

ACTUAL INPUT



QUALITY ASSESSMENT

Output

GLARE

PEREPRO-CESSING

selected representation

GRAY

DETECTION



PP-OCRv6 Medium
retained OCR boxes from selected GRAY artifact.

RECOGNITION

SVTRv2 Server
recognized text

OCR EVIDENCE EXCEPT

650211-11-5049
HAMZAH BIN KAMMAPU
PT 1160 P
JALAN KENANGA
20400 KUALA
TERENGGANU
TERENGGANU

Overall OCR confidence
0.9855

PREPROCESSING ADAPTS TO OTHER CONDITIONS

GRAY: Clean, Glare ← current

RAW: Low contrast, Slight blur

GRAY+CLAHE: Dark, Severe blur, Uneven lighting.

06 - RUNNING EXAMPLE

From OCR Evidence to the Right Specialist

The OCR output is interpreted as document evidence, then routed to the specialist extractor that matches its structure.

EVIDENCE DRIVEN ROUTING

1 OCR EVIDENCE
from slide 05

650211-11-5049
HAMZAH BIN KAMMAPU
PT 1160 P
JALAN KENANGA
20400 KUALA
TERENGGANU
TERENGGANU

2 ROUTING CUES
(current case)

MyKad Keyword
"MyKad"

Malaysian ID layout/keywords
e.g., KAD PENGENALAN

No MRZ structure detected.

3 COARSE ROUTE DECISION

MYKAD

CURRENT CASE

Assigned to the MYKAD coarse family

4 HOW OTHER DOCUMENTS ROUTE

MRZ evidence present
→ **MRZ_DOCUMENT**
→ MRZ / Passport Specialist

→ **GENERIC_ID**
→ Secondary country/layout routing.
→ EU DL/Country-Specific Specialist

No stable pattern matched
→ **UKNOWN_ROUTE**
→ Conservative fallback

5 CURRENT SPECIALIST
(running case)

MyKad Specialist

- Name candidate ranking (1-2 line candidates, RandomForest ranker)
- DOB interpretation from MyKad ID Number
- Address line grouping/ formatting

COARSE FAMILY DISTRIBUTION - LABELED SET (632 DOCUMENTS)

MYKAD
100
documents

MRZ_DOCUMENT
182
documents

GENERIC_ID
350
documents

GENERIC_ID is a routing bucket, not a real-world document type. It receives secondary country/layout routing after coarse-family assignment.

FROM MyKAD Specialist to Structured Fields

Field traceability from OCR evidence to final structured KYC fields.



Quality	Preprocessing
GLARE	GRAY
Route	**OCR Confidence**
MyKad	0.9855

FIELD TRACEABILITY

	OCR EVIDENCE	MYKAD SPECIALIST LOGIC	STRUCTURED OUTPUT
Name	HAMZAH BIN KAMMAPU	• MyKAD Name Specialist • 1-2 line candidates • RandomForest candidate ranking	HAMZAH BIN KAMMAPU
DOB	650211-11-5049	• First 6 digits as YYMMDD • 650211 → 1965-02-11	1965-02-11
ADDRESS	PT 1160 P JALAN KENANGA 20400 KUALA TERENGGANU TERENGGANU	• Reading-order grouping • Format normalization	PT 1160 P, JALAN KENANGA, 20400 KUALA TERENGGANU, TERENGGANU

Repeated Captures Are a Validation Risk and a Peer Evidence Opportunity

Repeated label-derived groups are not independent samples; they create leakage risk and provide a separate setting for cross-capture analysis.

1 WHAT MIXED CORRECTNESS LOOKS LIKE

ILLUSTRATIVE PATTERN label derived repeated group

	Capture A	Capture B	Capture C
Name	✓	✗	✓
DOB	✓	✓	✗

Different captures can preserve different fields.

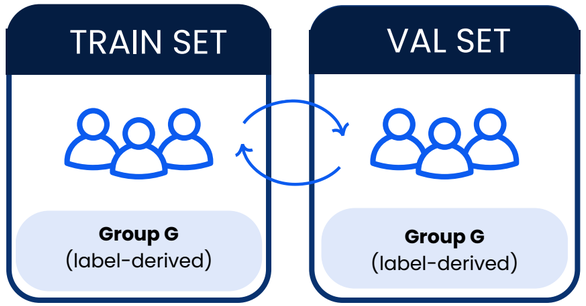
26 repeated groups

largest group: **40** captures

label-derived groups, not verified unique people

2 VALIDATION RISK

Same label-derived group can land in both train and validation sets.

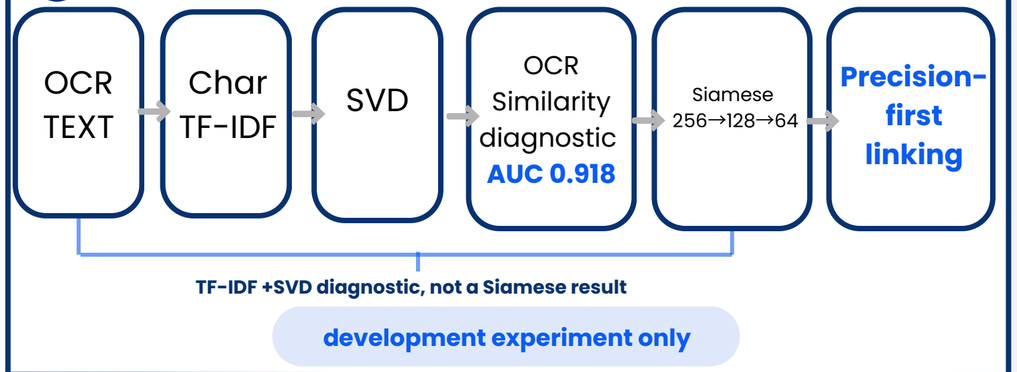


Leakage risk: performance can be overly optimistic when the sae group appears in both sets.

Use group-aware validation where applicable.

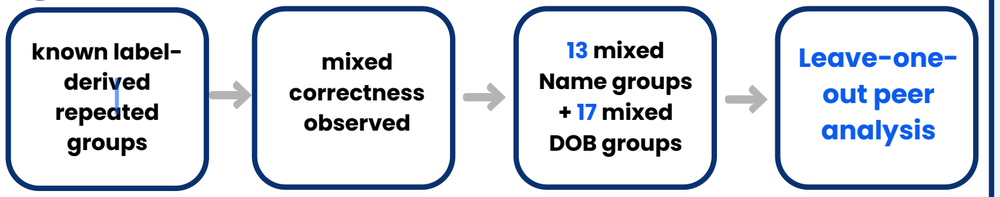
3 CROSS-CAPTURE ANALYSIS

A DEVELOPMENT IDENTITY-LINKING EXPERIMENT



development experiment only

B DOWNSTREAM LABELED PEER ANALYSIS



separate experiment only

TRUTHFUL SCOPE Siamese linking was a development experiment; LOO peer analysis uses known label-derived groups and is separate from core inference.

FINAL V28 Reaches 66.77% Macro Field EM, with Unknown-Route Names as the Main Bottleneck

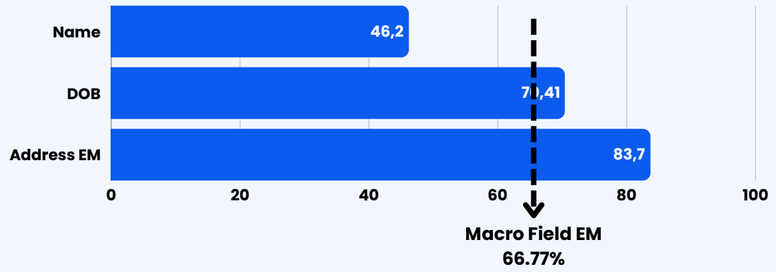
The 632-row V28 labeled benchmark shows uneven performance across fields: Address all-row EM is high but blank-sensitive, while UNKNOWN_ROUTE Name extraction remains the clearest bottleneck.

FINAL V28 BENCHMARK

66.77%
Macro Field EM

37.66%	81.28%	1.3495
Full Record EM	Normalized Similarity	Record CER (lower is better)

Field-Level Exact Match



Address caveat: 83.7% all-row EM; only 112 rowns contain valid Address labels; present-only EM = 25%.

PRIMARY BOTTLENECK

UNKNOWN_ROUTE fallback - 147 Generic_ID documents.

Name: 0% EM | 28.22% similarity

DOB: 19.73% EM | 41.16% similarity

The exact ground-truth Name appears in retained OCR text for only 1/147 documents.

ROOT CAUSES

- OCR recognition gap
- Unknown-layout fallback
- language/ script variability

IMPROVEMENT PATH

1. Broader route/layout coverage
2. Stronger Name OCR + recognition.
3. Better multilingual/unseen-layout handling.

Expected impact:
higher Name EM on unknown routes → higher Macro Field EM.

BENCHMARK CONTEXT | 632 labeled rows | field + record metrics | route-level error analysis | Group-aware validation is used separately for supervised model selection where applicable.

Automate the Repetitive Work, Escalate Uncertain Cases

Reduce repetitive identitiy-data extraction while keeping uncertain cases reviewable and auditable.

QUALITY GATE + GUIDED RETAKE

Prevent avoidable OCR failures before inference.

GLARE | BLUR | CROP | LOW LIGHT

Guided Retake

METADATA-ASSISTED ROUTING

Auto-route when evidence is clear; request country/document type only when routing is ambigous.

OCR Evidence

↓

Routing cues/ ambiguity check

↓

Optional country/ document type input

EVIDENCE BASED SELECTIVE REVIEW

Auto-prefill supported cases; send weak or conflicting evidence to reviewers.

SUPPORTED EVIDENCE → **Auto-prefilled fields**

WEAK/ CONFLICTING/ UNKNOWN ROUTE → **Human Review**

FIELD MONITORING + AUDITABLE REVIEW

Monitor field quality and give reviewers the evidence behind every extracted value.

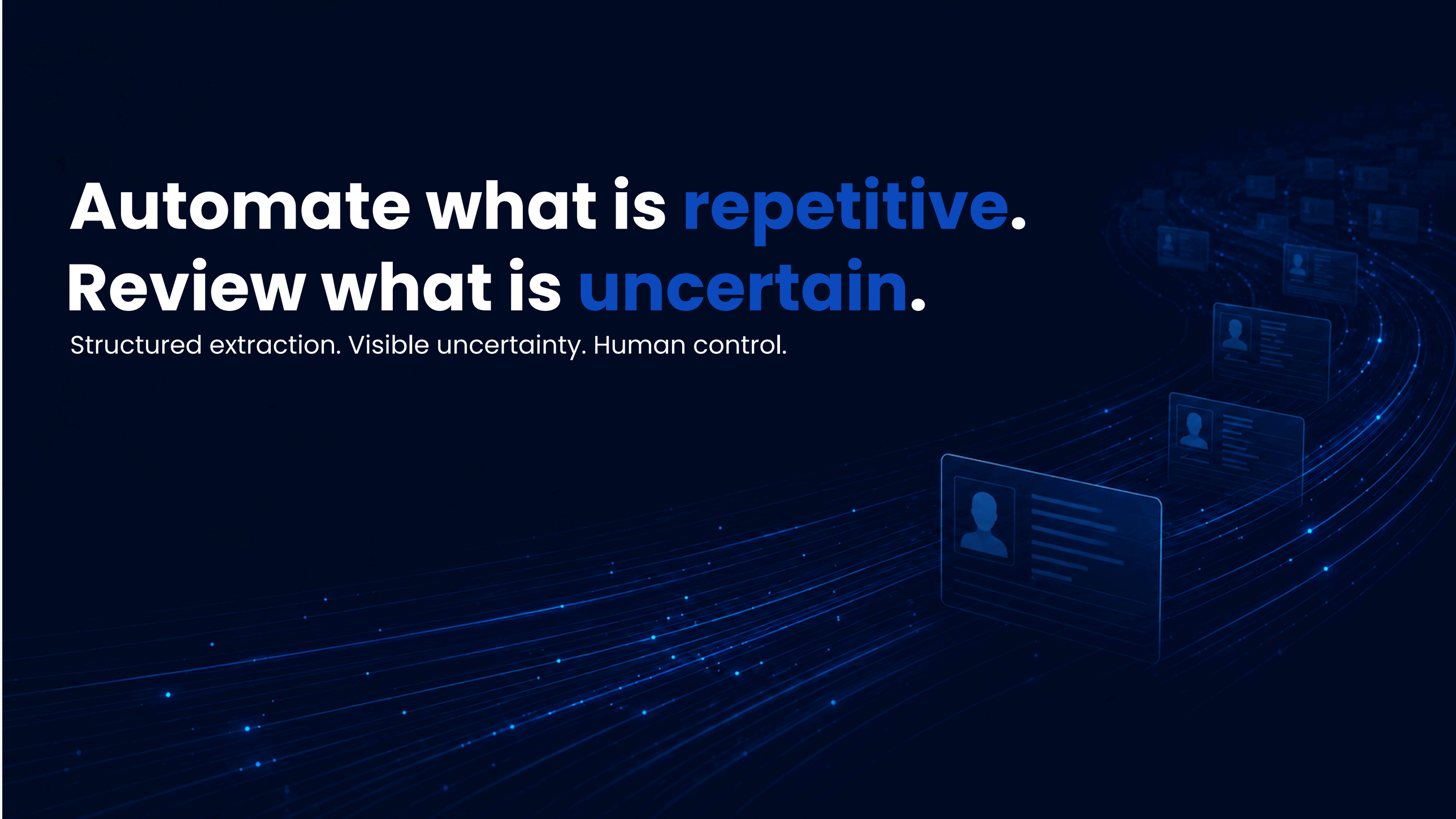
- Processed volume
- Flagged review cases
- Field error trend
- BBox/ evidence trace

Illustrative operations dashboard - not measured results

FUTURE R&D ● Verified repeated-capture reconciliation only after reliable identitiy linking.

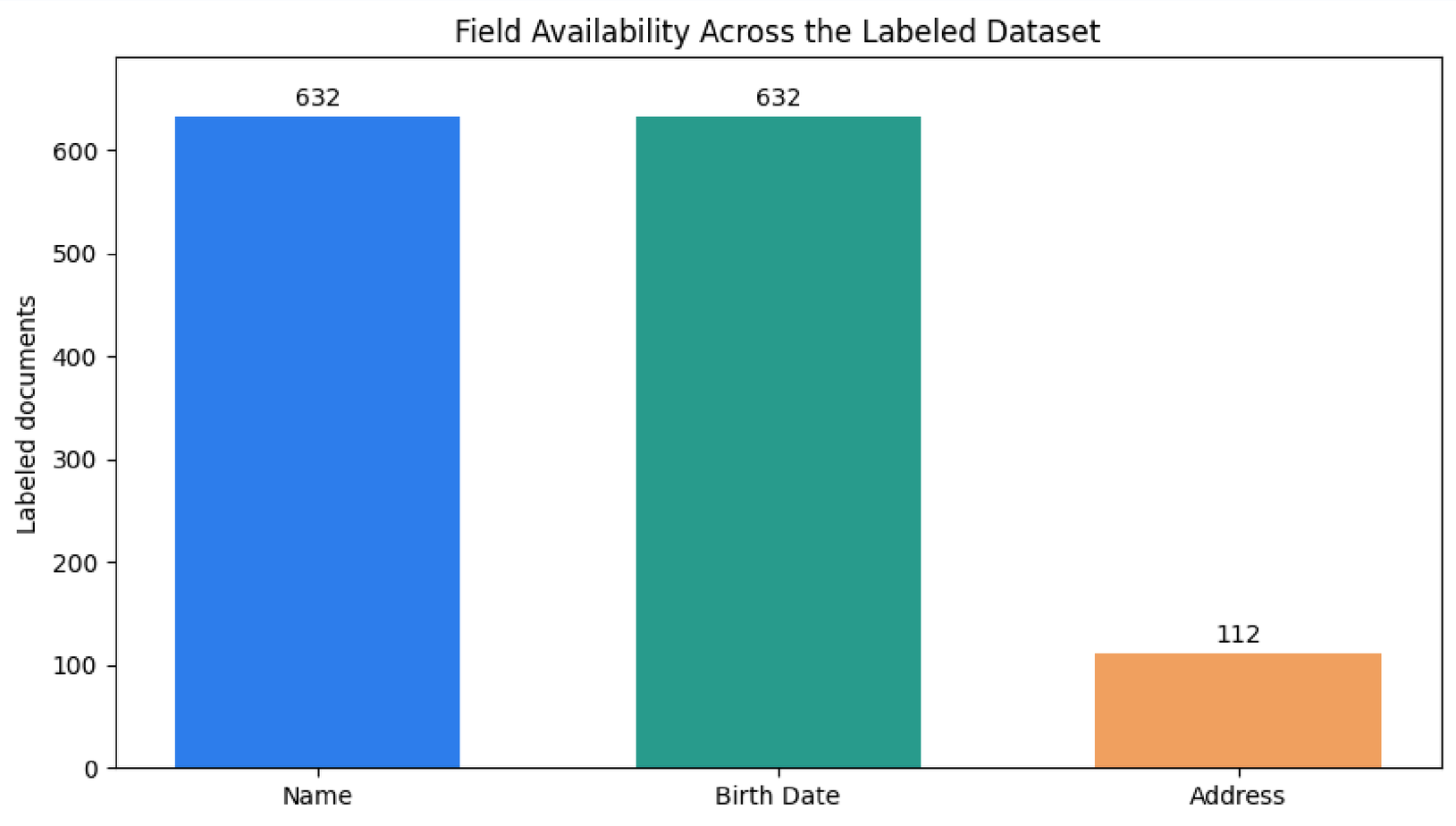
Automate what is repetitive.
Review what is uncertain.

Structured extraction. Visible uncertainty. Human control.



APPENDIX

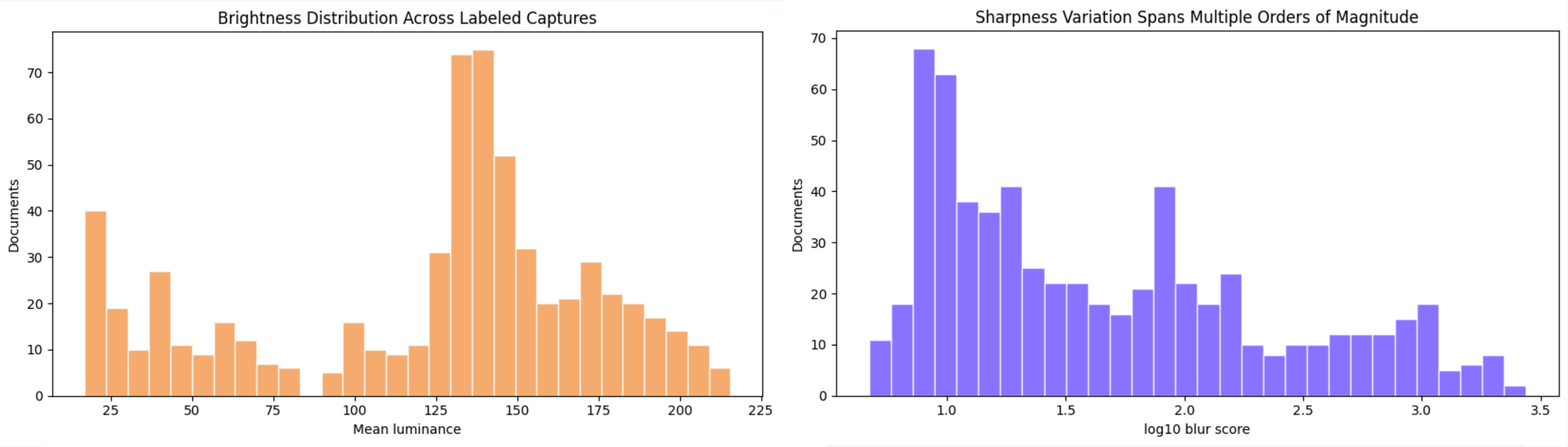
A1. Label Distribution

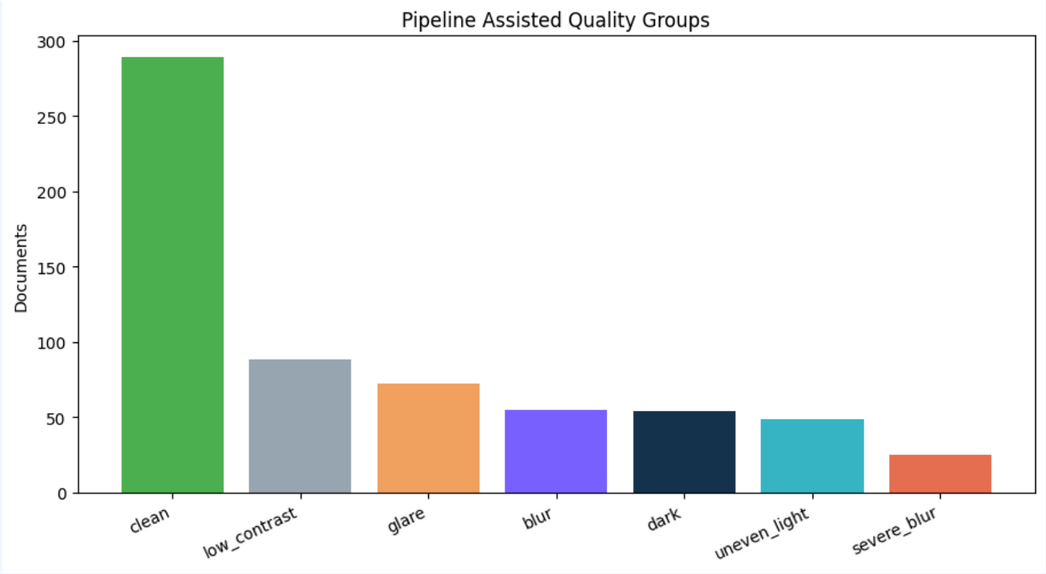


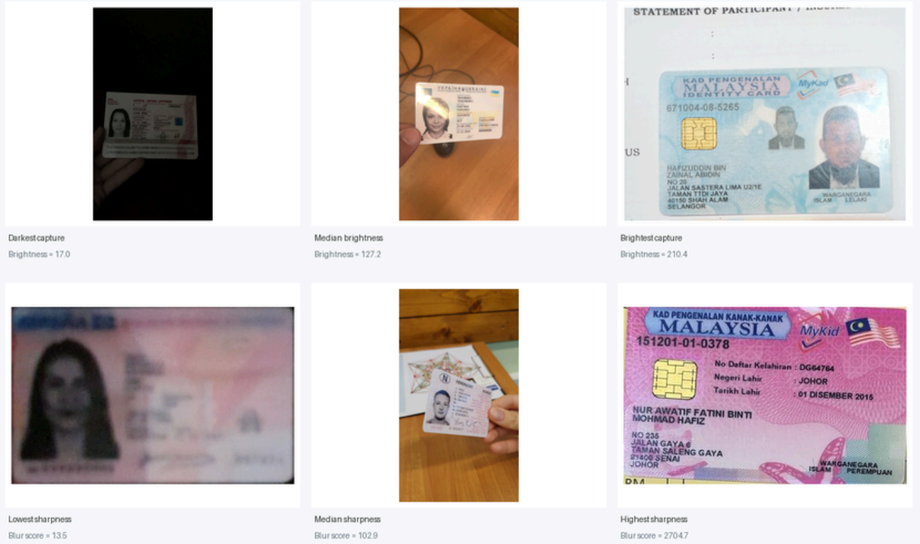
Evaluation Scope	Rows	Address Exact Match	Interpretation
All rows	632	83.70%	Blank-sensitive; includes rows with no operationally valid Address GT
Address present only	112	25.00%	Actual exact-match performance on rows where Address exists

```
Address-present share = 17.7%
Defense: Address is NOT considered solved based on the all-row score alone.
```

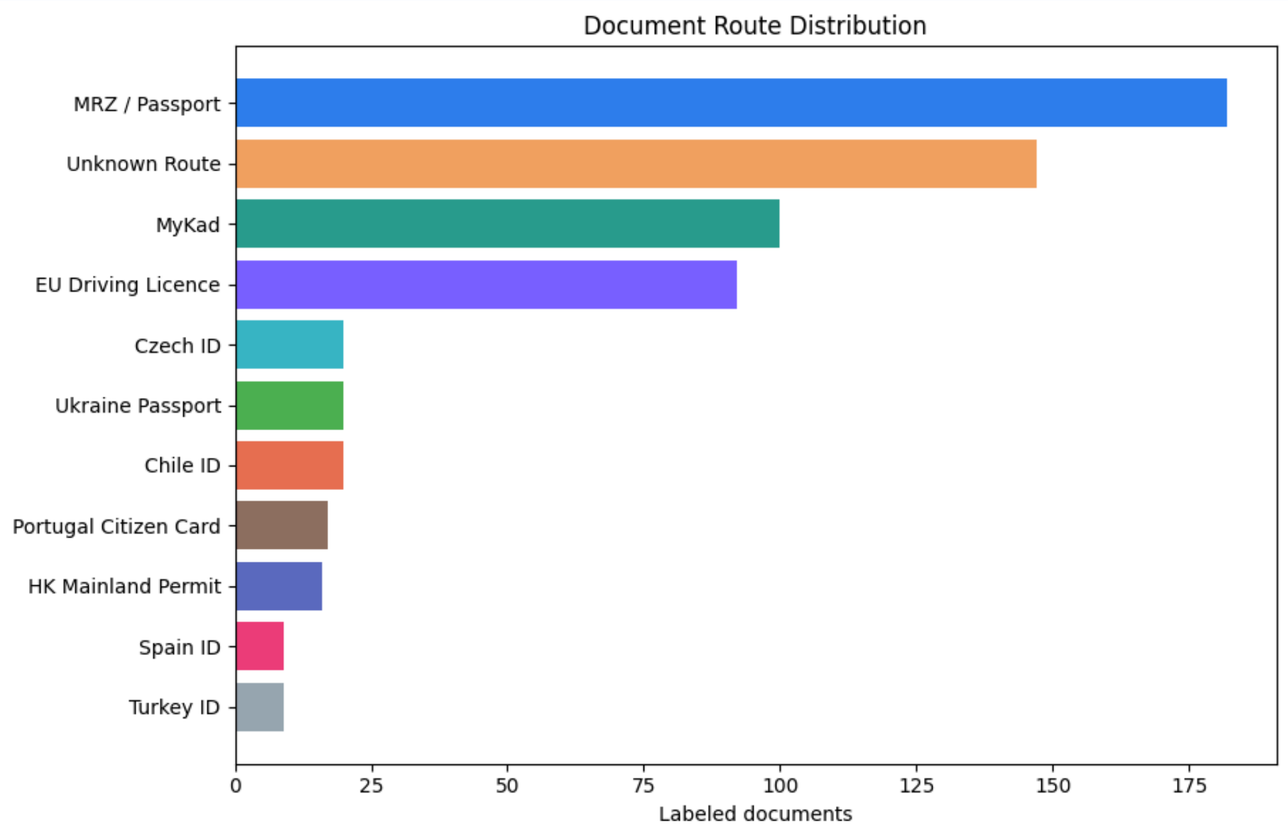
A2. Dataset Quality Audit







A3. Document Route





Mrz / Passport

image_125.jpg | 27 OCR boxes



Unknown Route

image_062.jpg | 11 OCR boxes



Mykad

image_003.jpg | 14 OCR boxes



Eu Driving License

image_218.jpg | 16 OCR boxes



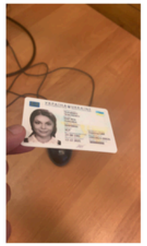
Chile Id

image_134.jpg | 23 OCR boxes



Czech Id

image_175.jpg | 22 OCR boxes



Ukraine Passport

image_595.jpg | 23 OCR boxes



Portugal Citizen Card

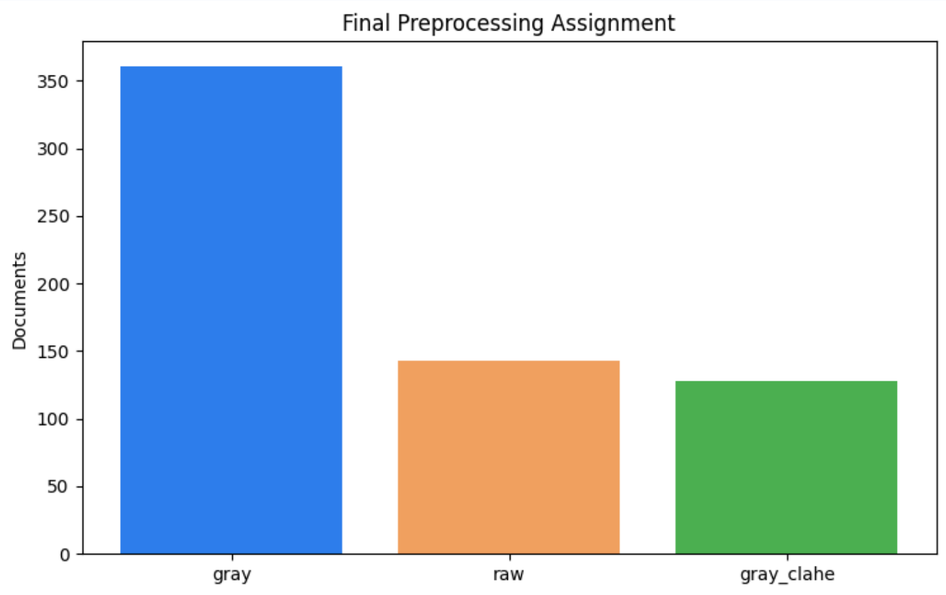
image_554.jpg | 10 OCR boxes



Hk Mainland Permit

image_153.jpg | 14 OCR boxes

A4. Preprocessing Distribution



Quality Group	Selected Preprocessing	Documents
blur	raw	55
clean	gray	289
dark	gray_clahe	54
glare	gray	72
low_contrast	raw	88
severe_blur	gray_clahe	25
uneven_light	gray_clahe	49

Selected Preprocessing	Documents
gray	361
raw	143
gray_clahe	128



A5. OCR Transcript

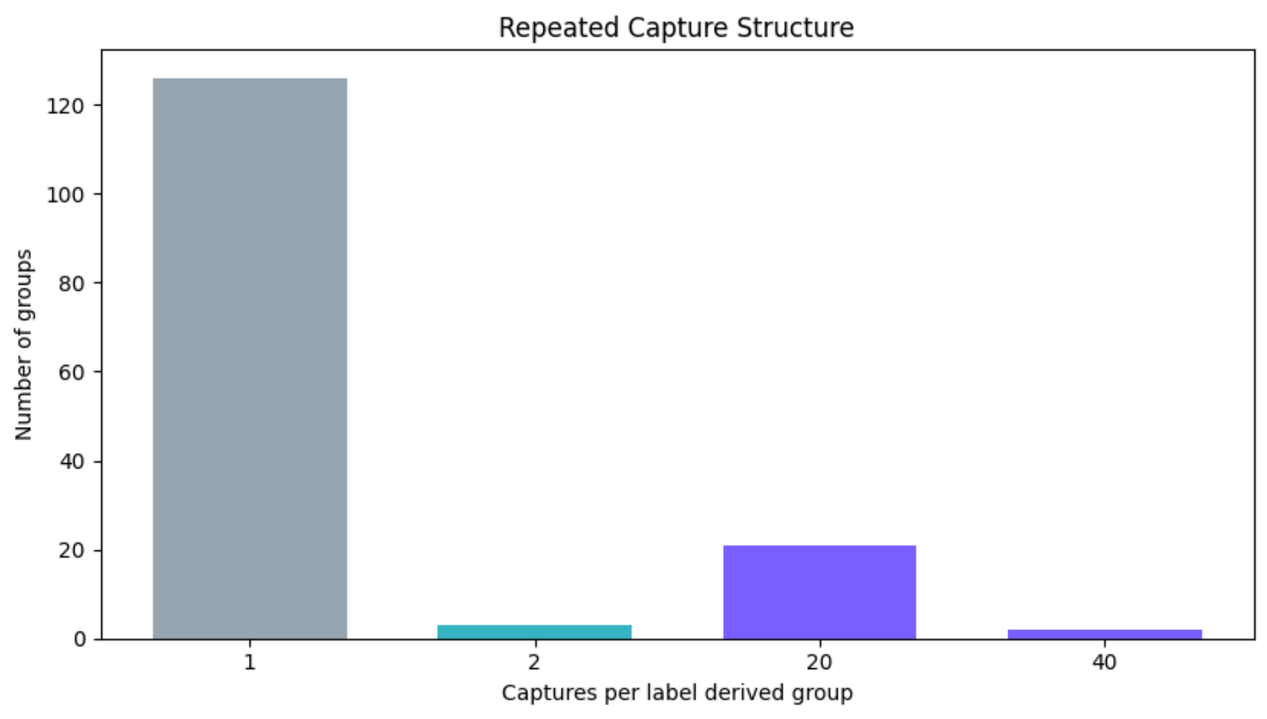
image_001.jpg — rectified document

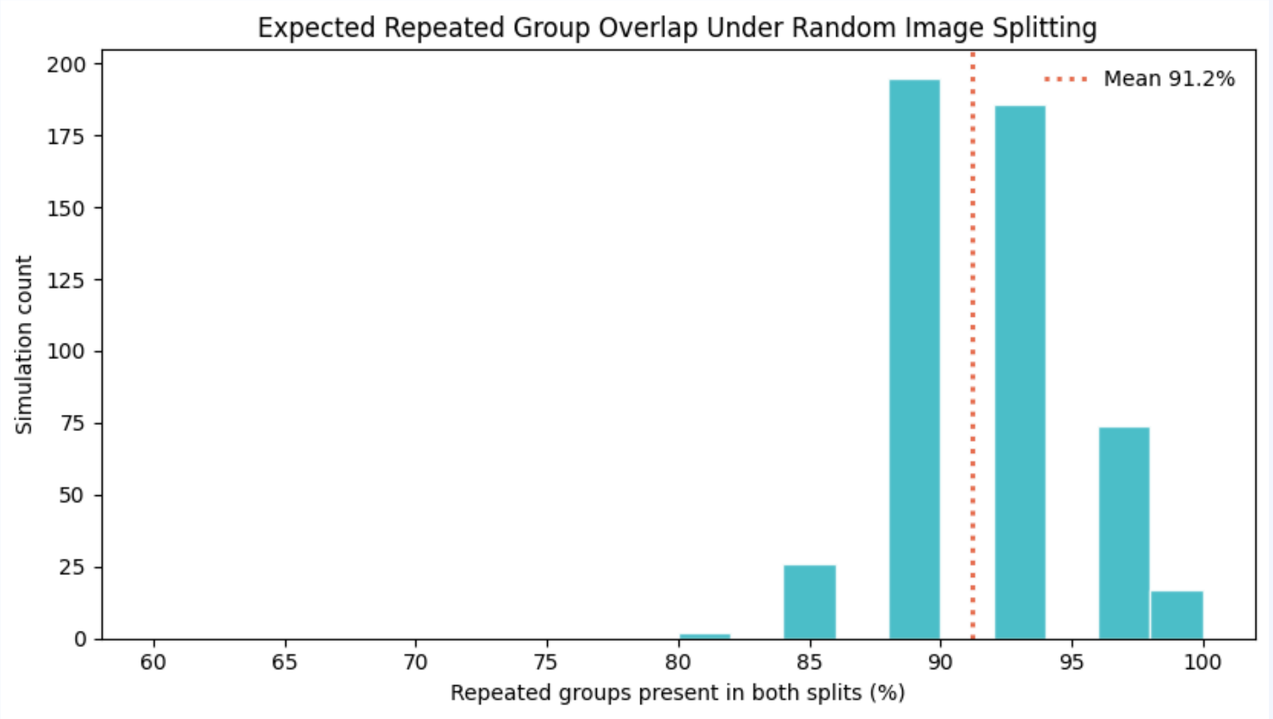


Frozen final OCR transcript

```
KAD PENGENALAN
MyKad
MALAYSIA
650211-11-5049
HAMZAH BIN KAMMAPU
PT1160P
JALAN KENANGA
20400 KUALA TERENGGANU
WARGANEGARA
TERENGGANU
ISLAM
LELAKI
```

A6.Splitting





A7. Mixed Repeated-Capture Group Example

Example mixed group: id_ffb4dd8d8038

filename	name_gt	name_pred	name_exact	birth_date_gt	birth_date_pred	dob_exact	ocr_confidence	quality_group
image_213.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.935489	clean
image_214.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.924174	clean
image_215.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.928689	clean
image_216.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.923753	clean
image_217.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.930679	clean
image_218.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.938583	glare
image_219.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.931624	glare
image_220.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.925577	glare
image_221.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.812080	clean
image_222.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.931399	glare
image_223.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.859739	low_contrast
image_224.jpg	ERIKA MUSTERMANN	Musterhuusen am Sro	False	1964-08-12	nan	False	0.558226	low_contrast
image_225.jpg	ERIKA MUSTERMANN	Musterhuusen am Sro	False	1964-08-12	nan	False	0.548243	low_contrast
image_226.jpg	ERIKA MUSTERMANN	Musterhuusen am Sro	False	1964-08-12	nan	False	0.685412	low_contrast
image_227.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.806591	low_contrast
image_228.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.873665	low_contrast
image_229.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.928578	low_contrast
image_230.jpg	ERIKA MUSTERMANN	Erika Mustermann	True	1964-08-12	1964-08-12	True	0.800989	low_contrast
image_231.jpg	ERIKA MUSTERMANN	Musterhuusen am Sro	False	1964-08-12	nan	False	0.838654	low_contrast
image_232.jpg	ERIKA MUSTERMANN	Musterrausen ah see	False	1964-08-12	nan	False	0.838617	low_contrast

A8. Metric Formula

1 Name Exact Match

$$\text{Name EM} = \frac{\#\text{ documents with exact Name prediction}}{N}$$

Exact string match on Name field.

2 DOB Exact Match

$$\text{DOB EM} = \frac{\#\text{ documents with exact DOB prediction}}{N}$$

Exact match after DOB normalization.

3 Address Exact Match

$$\text{Address EM} = \frac{\#\text{ documents with exact Address prediction}}{N}$$

All-row metric; blank-sensitive.

4 Macro Field EM

$$\text{Macro Field EM} = \frac{\text{Name EM} + \text{DOB EM} + \text{Address EM}}{3}$$

Gives equal weight to the three target fields.

5 Full Record EM

$$\text{Full Record EM} = \frac{\#\text{ documents where Name, DOB, and Address are all correct}}{N}$$

A stricter whole-record metric.

6 Record CER

$$\text{CER} = \frac{S + D + I}{N_{chars}}$$

S = substitutions, D = deletions, I = insertions

Character-level error severity; lower is better.

7 Normalized Edit Similarity (NES)

$$\text{NES} = 1 - \frac{S + D + I}{\max(|s_{ref}|, |s_{pred}|)}$$

S = substitutions
D = deletions
I = insertions
s_{ref} = reference string
s_{pred} = predicted string
$|s|$ = length of string s

Range: [0, 1] | 1 = exact match | 0 = completely different

Length-normalized Levenshtein similarity; robust to length differences.

A9. Runtime Configuration

Stage	Model	Runtime	Role
Text Detection	PP-OCRv6 Medium	ONNX / CPU	Produces OCR bounding boxes / geometry
Text Recognition	OpenOCR SVTRv2 Server	PyTorch CUDA / A100	Recognizes detected text regions
Combined runtime	RapidOCR detector + OpenOCR recognizer	CPU detector + GPU recognizer	Retained V28 OCR configuration

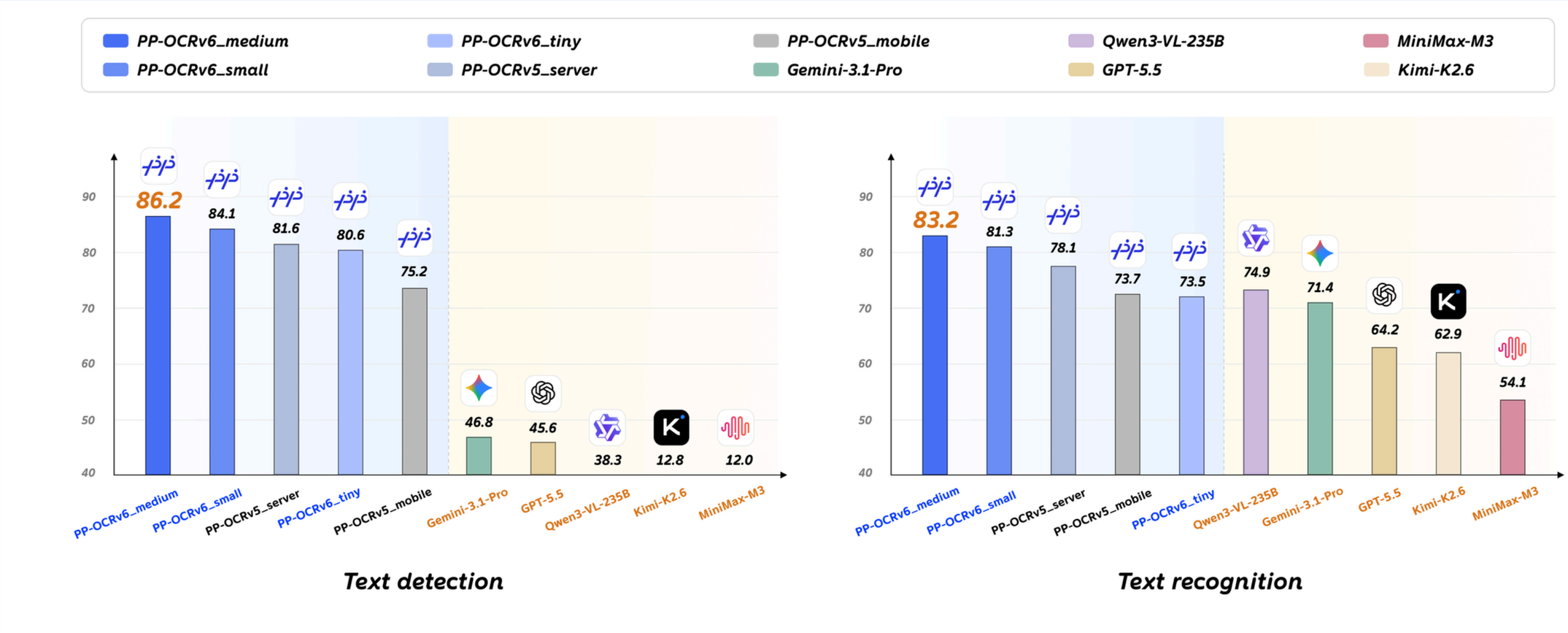
Configuration Item	Retained Value
ocr_model	ppocrv6_medium_det_cpu__svtrv2_server_rec_gpu
recognizer checkpoint	OpenOCR-0d52280-SVTRv2-server-3be361b
runtime	rapidocr_onnxruntime_cpu_detector + openocr_torch_cuda_recognizer
configuration	PP-OCRv6 Medium (CPU detector) + SVTRv2 Server (A100 recognizer) / quality-aware adaptive

A10. Unknown-Route

Route	Documents	Name EM	Name Similarity	DOB EM
Unknown Route	147	0.00%	28.22%	19.73%
Portugal Citizen Card	17	0.00%	56.25%	100.00%
MRZ / Passport	182	46.70%	70.23%	71.43%
Chile ID	20	0.00%	91.67%	100.00%
Spain ID	9	0.00%	91.67%	100.00%
MyKad	100	63.00%	95.48%	100.00%
Czech ID	20	95.00%	96.05%	100.00%
EU Driving Licence	92	86.96%	98.40%	81.52%
HK Mainland Permit	16	100.00%	100.00%	100.00%
Turkey ID	9	100.00%	100.00%	100.00%
Ukraine Passport	20	100.00%	100.00%	100.00%

Unknown-Route Diagnostic	Value
Unknown Route documents	147
Name Exact Match	0.00%
Name normalized similarity	28.22%
DOB Exact Match	19.73%
Exact GT Name found in retained OCR text	1 / 147 documents

A11. Why Paddle OCR



A12. Why SVTRv2

				IIIT5k	*SVT*	*ICDAR2013*	*ICDAR2015*	*SVTP*	*CUTE80*		*Curve*	*Multi-Oriented*	*Artistic*	*Contextless*	*Salient*	*Multi-Words*	*General*					
Method		Venue	Encoder	Common Benchmarks (*Com*)						Avg	Union14M-Benchmark (*U14M*)							Avg	*LTB*	*OST*	*Size*	*FPS*
ASTER [40]		TPAMI19	ResNet+LSTM	96.1	93.0	94.9	86.1	87.9	92.0	91.70	70.9	82.2	56.7	62.9	73.9	58.5	76.3	68.75	0.1	61.9	19.0	67.1
NRTR [38]		ICDAR19	Stem+TF_6	98.1	96.8	97.8	88.9	93.3	94.4	94.89	67.9	42.4	66.5	73.6	66.4	77.2	78.3	67.46	0.0	74.8	44.3	17.3
MORAN [32]		PR19	ResNet+LSTM	96.7	91.7	94.6	84.6	85.7	90.3	90.61	51.2	15.5	51.3	61.2	43.2	64.1	69.3	50.82	0.1	57.9	17.4	59.5
SAR [29]		AAAI19	ResNet+LSTM	98.1	93.8	96.7	86.0	87.9	95.5	93.01	70.5	51.8	63.7	73.9	64.0	79.1	75.5	68.36	0.0	60.6	57.5	15.8
DAN [46]		AAAI20	ResNet+FPN	97.5	94.7	96.5	87.1	89.1	94.4	93.24	74.9	63.3	63.4	70.6	70.2	71.1	76.8	70.05	0.0	61.8	27.7	99.0
SRN [55]		CVPR20	ResNet+FPN	97.2	96.3	97.5	87.9	90.9	96.9	94.45	78.1	63.2	66.3	65.3	71.4	58.3	76.5	68.43	0.0	64.6	51.7	67.1
SEED [36]		CVPR20	ResNet+LSTM	96.5	93.2	94.2	87.5	88.7	93.4	92.24	69.1	80.9	56.9	63.9	73.4	61.3	76.5	68.87	0.1	62.6	24.0	65.4
AutoSTR [59]		ECCV20	NAS+LSTM	96.8	92.4	95.7	86.6	88.2	93.4	92.19	72.1	81.7	56.7	64.8	75.4	64.0	75.9	70.09	0.1	61.5	6.0	82.6
RoScanner [57]		ECCV20	ResNet	98.5	95.8	97.7	88.2	90.1	97.6	94.65	79.4	68.1	70.5	79.6	71.6	82.5	80.8	76.08	0.0	68.6	48.0	64.1
ABINet [15]		CVPR21	ResNet+TF_3	98.5	98.1	97.7	90.1	94.1	96.5	95.83	80.4	69.0	71.7	74.7	77.6	76.8	79.8	75.72	0.0	75.0	36.9	73.0
VisionLAN [47]		ICCV21	ResNet+TF_3	98.2	95.8	97.1	88.6	91.2	96.2	94.50	79.6	71.4	67.9	73.7	76.1	73.9	79.1	74.53	0.0	66.4	32.9	93.5
PARSeq [4]		ECCV22	ViT-S	98.9	98.1	98.4	90.1	94.3	98.6	96.40	87.6	88.8	76.5	83.4	84.4	84.3	84.9	84.26	0.0	79.9	23.8	52.6
MATRN [34]		ECCV22	ResNet+TF_3	98.8	98.3	97.9	90.3	95.2	97.2	96.29	82.2	73.0	73.4	76.9	79.4	77.4	81.0	77.62	0.0	77.8	44.3	46.9
MGP-STR [45]		ECCV22	ViT-B	97.9	97.8	97.1	89.6	95.2	96.9	95.75	85.2	83.7	72.6	75.1	79.8	71.1	83.1	78.65	0.0	78.7	148	120
LPV [58]		IJCAI23	SVTR-B	98.6	97.8	98.1	89.8	93.6	97.6	95.93	86.2	78.7	75.8	80.2	82.9	81.6	82.9	81.20	0.0	77.7	30.5	82.6
MAERec [25]		ICCV23	ViT-S	**99.2**	97.8	98.2	90.4	94.3	98.3	96.36	89.1	87.1	79.0	84.2	**86.3**	85.9	84.6	85.17	9.8	76.4	35.7	17.1
LISTER [8]		ICCV23	FocalNet-B	98.8	97.5	98.6	90.0	94.4	96.9	96.03	78.7	68.8	73.7	81.6	74.8	82.4	83.5	77.64	36.3	77.1	51.1	44.6
CDistNet [65]		IJCV24	ResNet+TF_3	98.7	97.1	97.8	89.6	93.5	96.9	95.59	81.7	77.1	72.6	78.2	79.9	79.7	81.1	78.62	0.0	71.8	43.3	15.9
CAM [54]		PR24	ConvNeXtV2	98.2	96.1	96.6	89.0	93.5	96.2	94.94	85.4	89.0	72.0	75.4	84.0	74.8	83.1	80.52	0.7	74.2	58.7	28.6
BUSNet [49]		AAAI24	ViT-S	98.3	98.1	97.8	90.2	**95.3**	96.5	96.06	83.0	82.3	70.8	77.9	78.8	71.2	82.6	78.10	0.0	78.7	32.1	83.3
OTE [52]		CVPR24	SVTR-B	98.6	96.6	98.0	90.1	94.0	97.2	95.74	86.0	75.8	74.6	74.7	81.0	65.3	82.3	77.09	0.0	77.8	20.3	55.2
CPPD [13]		TPAMI25	SVTR-B	99.0	97.8	98.2	90.4	94.0	**99.0**	96.40	86.2	78.7	76.5	82.9	83.5	81.9	83.5	81.91	0.0	79.6	27.0	125
IGTR-AR [14]		TPAMI25	SVTR-B	98.7	**98.4**	98.1	90.5	94.9	98.3	96.48	90.4	**91.2**	77.0	82.4	84.7	84.0	84.4	84.86	0.0	76.3	24.1	58.3
SMTR [12]		AAAI25	FocalSVTR	99.0	97.4	98.3	90.1	92.7	97.9	95.90	89.1	87.7	76.8	83.9	84.6	**89.3**	83.7	85.00	**55.5**	73.5	15.8	66.2
C	CRNN [39]	TPAMI16	ResNet+LSTM	95.8	91.8	94.6	84.9	83.1	91.0	90.21	48.1	13.0	51.2	62.3	41.4	60.4	68.2	49.24	47.2	58.0	16.2	172
C	SVTR [11]	IJCAI22	SVTR-B	98.0	97.1	97.3	88.6	90.7	95.8	94.58	76.2	44.5	67.8	78.7	75.2	77.9	77.8	71.17	45.1	69.6	18.1	161
T	SVTRv2	-	SVTRv2-T	98.6	96.6	98.0	88.4	90.5	96.5	94.78	83.6	76.0	71.2	82.4	77.2	82.3	80.7	79.05	47.8	71.4	5.1	201
C			SVTRv2-S	99.0	98.3	98.5	89.5	92.9	98.6	96.13	88.3	84.6	76.5	84.3	83.3	85.4	83.5	83.70	47.6	78.0	11.3	189
			SVTRv2-B	**99.2**	98.0	**98.7**	**91.1**	93.5	**99.0**	**96.57**	**90.6**	89.0	**79.3**	**86.1**	86.2	86.7	**85.1**	**86.14**	50.2	**80.0**	19.8	143

Table 3. All the models and SVTRv2 are trained on *U14M-Filter*. To ensuring that the results reflect the true potential of these methods under their best experimental settings, we conducted extensive tuning (detailed in *Suppl. Sec.* 12) of the model-specific settings (e.g., optimizer, learning rate, and regularization) and reported the best result we got. TF_n denotes the n-layer Transformer block [41]. *Size* denotes the number of parameters of the model ($\times 10^6$). *FPS* is measured on one NVIDIA 1080Ti GPU.